FinRAG: Financial Retrieval-Augmented Generation System

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1. Project Title

A Unified AI Framework for Financial Risk and Retrieval-Augmented Intelligence

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2. Real-World Problem

Problem: Financial analysts, investors, and compliance officers spend 15-20 hours/week manually searching through lengthy SEC filings (10-K, 10-Q reports), annual reports, and financial statements to answer complex queries like "What was Apple's Q3 2024 revenue compared to Q3 2023?" or "List all risk factors related to supply chain disruptions in Tesla's 2024 10-K."

Current Limitations:

- Traditional keyword search fails on complex financial queries requiring multi-hop reasoning
- LLMs hallucinate when answering without grounding in actual documents
- Existing RAG systems achieve only 19-43% accuracy on financial QA benchmarks (FinanceBench)computer+2
- Financial documents contain dense tables, technical terminology, and cross-referenced data that standard RAG cannot handle

Who Benefits:

- **Financial analysts** (reduce research time by 70%)
- **Retail investors** (access institutional-grade research)
- **Compliance officers** (verify regulatory filings quickly)
- **Investment advisors** (answer client queries with citations)

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3. Proposed AI Solution

FinRAG-Enhanced is a web-based financial QA system that:

1. Ingests financial documents (10-K, 10-Q, annual reports)
2. Creates metadata-rich document chunks using LLM-generated summaries
3. Uses hybrid search (dense + sparse + metadata filtering) for retrieval
4. Applies LLM-based query expansion (HyDE) to improve retrieval coverage
5. Reranks retrieved chunks with cross-encoder
6. Generates accurate answers with citations using LLM (Llama 2/Mistral)

Key Features:

- Answer complex financial questions with 85%+ accuracy (vs 19-43% baseline)
- Provide source citations (page numbers, document sections)
- Handle tabular data (financial statements, balance sheets)
- Support multi-hop queries (e.g., "Compare revenue growth of Apple vs Microsoft over 3 years")

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4. Base Research Paper ✅ DOWNLOADABLE

Primary Base Paper:

Paper Title: Metadata-Driven Retrieval-Augmented Generation for Financial Question Answering

Authors: Michail Dadopoulos, Anestis Ladas, Stratos Moschidis, Ioannis Negkakis

Year: 2025 (submitted October 28, 2025)

Journal/Conference: arXiv preprint

arXiv ID: 2510.24402

DOI/Link: https://arxiv.org/abs/2510.24402

PDF Download: Click "View PDF" on arXiv page (FREE, no login required)arxiv

Dataset Used:

- **FinanceBench:** 1,000+ financial QA pairs (public benchmark)
- **FinDER:** 5,703 expert-annotated query-evidence-answer tripletsarxiv

- **SEC EDGAR Filings:** 10-K, 10-Q reports (public, millions of documents)

Algorithms/Models Used:

- LLM-generated metadata (GPT-3.5/4 for chunk summaries)
- Dense embeddings (text-embedding-004, all-mpnet-base-v2)
- Hybrid search (dense + BM25 sparse retrieval)
- Cross-encoder reranking (ms-marco-T5-base)
- LLM for answer generation (Gemini-2.0, Llama 2)lib.uom

Secondary Reference Papers:

Paper 2: FinSage: A Multi-aspect RAG System for Financial Filings Question Answering

- **arXiv:** 2504.14493 (FREE)
- **Link:** https://arxiv.org/abs/2504.14493
- **Contribution:** Multi-modal RAG for financial filings, 92.51% recallarxiv

Paper 3: Enhancing Financial RAG with Agentic AI and Multi-HyDE

- **arXiv:** 2509.16369 (FREE)
- **Link:** https://arxiv.org/abs/2509.16369
- **Contribution:** Agentic RAG with Multi-HyDE query expansion, 11.2% accuracy improvementarxiv

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5. Research Gap

The base paper achieves 88% accuracy but has critical limitations:lib.uom

1. **Limited to FinanceBench:** Evaluated only on FinanceBench (1,000 questions), not on real-world enterprise financial documents (annual reports, earnings call transcripts)
2. **No Multi-Hop Reasoning:** Cannot handle complex queries requiring multi-step reasoning (e.g., "Calculate 3-year CAGR of Apple's revenue and compare with industry average")
3. **No Table Handling:** Financial statements (balance sheets, income statements) are in tables; base paper uses text-only chunking, losing table structure

4. **No Real-Time Updates:** Static document corpus; cannot ingest real-time filings (new 10-K/10-Q as soon as filed)
5. **No Citation Grounding:** Answers lack precise citations (page numbers, section names), reducing trust for compliance use
6. **No Indian Market Support:** Evaluated only on US SEC filings; no support for Indian financial documents (BSE/NSE filings, annual reports in Indian format)
7. **High Latency:** Multi-stage pipeline (metadata generation + hybrid search + reranking) adds 3-5 seconds latency per query

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6. Proposed Novelty

You will add these improvements over base paper:

1. **Table-Aware Chunking:** Preserve table structure in financial statements using HTML/Markdown chunking (base paper loses table context)
2. **Multi-Hop Query Support:** Implement query decomposition agent that breaks complex queries into sub-queries (e.g., "Calculate CAGR" → "Get revenue for 2021, 2022, 2023" → "Compute growth rate")
3. **Indian Market Extension:** Add BSE/NSE filings (Indian companies: Reliance, TCS, HDFC Bank) alongside SEC filings
4. **Real-Time Ingestion Pipeline:** Automated SEC EDGAR + BSE India API integration to ingest new filings within 1 hour of publication
5. **Precise Citation Grounding:** Add page numbers, section names, table row/column references in answers (e.g., "Revenue: $383B [10-K 2024, Page 23, Table 1]")
6. **Latency Optimization:** Model distillation for faster embeddings (MiniLM vs mpnet), caching for repeated queries, parallel retrieval
7. **HyDE Query Expansion:** Implement Hypothetical Document Embeddings (HyDE) to expand queries with LLM-generated hypothetical answersarxiv+1

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7. AI/ML Techniques

- **Retrieval-Augmented Generation (RAG):** Dense retrieval + LLM generation
- **Hybrid Search:** Dense embeddings (all-mpnet-base-v2) + Sparse (BM25)

- **LLM-Based Metadata Generation:** GPT-3.5/Llama 2 for chunk summaries
- **Cross-Encoder Reranking:** ms-marco-T5-base, BGE-reranker
- **Query Expansion:** HyDE (Hypothetical Document Embeddings), Multi-HyDEarxiv
- **Table Processing:** Table Transformer (TAPAS), HTML/Markdown chunking
- **Query Decomposition:** LLM agent for multi-hop reasoning
- **Named Entity Recognition (NER):** SpaCy FinBERT for company names, financial metrics
- **Model Distillation:** MiniLM for faster embeddings
- **Caching:** Redis for repeated query caching

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8. Dataset ✅ FREE & PUBLIC

Dataset 1: FinanceBench

- **Name:** FinanceBench
- **Source:** https://huggingface.co/datasets/rungalileo/ragbench (includes FinanceBench)
- **Number of Records:** 1,000+ financial QA pairs
- **Important Features:** Questions, ground truth answers, reference documents (10-K filings)
- **Whether Free:** Yes (HuggingFace, open access)arxiv

Dataset 2: FinDER

- **Name:** FinDER (Financial Dataset for RAG)
- **Source:** https://huggingface.co/datasets/Linq-AI-Research/FinDER
- **Number of Records:** 5,703 expert-annotated query-evidence-answer tripletsarxiv
- **Important Features:** Real-world financial queries, evidence spans, answers
- **Whether Free:** Yes (HuggingFace, open access)

Dataset 3: SEC EDGAR Filings

- **Name:** SEC EDGAR 10-K/10-Q Filings
- **Source:** https://www.sec.gov/edgar (official SEC website)

- **Number of Records:** Millions of filings (use subset: 100-500 companies, 2020-2024)
- **Important Features:** 10-K annual reports, 10-Q quarterly reports, financial statements, risk factors
- **Whether Free:** Yes (public, no login required)

Dataset 4: BSE/NSE India Filings (for Indian market extension)

- **Name:** BSE India Annual Reports
- **Source:** https://www.bseindia.com/ (BSE website) or https://github.com/kanishakbhatia/indian-stock-exchange
- **Number of Records:** 5,000+ Indian company filings (Reliance, TCS, HDFC, etc.)
- **Important Features:** Annual reports, quarterly results, balance sheets
- **Whether Free:** Yes (public)

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9. System Architecture

text

User (Financial Analyst) → React Web App (Frontend) →

FastAPI Backend →

[Query Processing] → NER (FinBERT) + Query Expansion (HyDE) →

[Retrieval] → Hybrid Search (Dense + BM25 + Metadata Filter) →

[Reranking] → Cross-Encoder (ms-marco-T5) →

[Answer Generation] → LLM (Llama 2/Mistral) →

PostgreSQL (Document Store) + Chroma/FAISS (Vector DB) →

Answer with Citations (Page Numbers, Sections)

Flow:

1. **User** submits query (e.g., "What was Apple's Q3 2024 revenue?")
2. **Frontend** (React) sends query to FastAPI backend
3. **Query Processing:**
    - NER extracts entities (company: Apple, period: Q3 2024, metric: revenue)

    - HyDE generates hypothetical answer for query expansion
4. **Retrieval:**
    - Dense search (all-mpnet-base-v2 embeddings)
    - Sparse search (BM25 keyword matching)
    - Metadata filter (company=Apple, year=2024, document_type=10-Q)
5. **Reranking:** Cross-encoder reranks top-50 chunks to top-10
6. **Answer Generation:** Llama 2 generates answer with citations
7. **Output:** "Apple's Q3 2024 revenue was $89.5B [10-Q 2024, Page 12, Table 2]"

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10. Technology Stack

Frontend:

- **Framework:** React.js + TypeScript
- **UI Components:** Material-UI or Tailwind CSS
- **Visualization:** Recharts (for financial charts), React-Table (for tabular data)
- **State Management:** Redux Toolkit or Zustand

Backend:

- **Framework:** Python FastAPI
- **API:** REST API for queries, document upload, answer retrieval
- **Task Queue:** Celery + Redis (for async document processing)

AI/ML:

- **Embeddings:** SentenceTransformers (all-mpnet-base-v2, MiniLM)
- **Reranking:** CrossEncoder (ms-marco-T5-base), BGE-reranker
- **LLM:** Llama 2 7B/13B (via HuggingFace or Ollama), Mistral 7B
- **NER:** SpaCy + FinBERT (financial entity extraction)
- **Query Expansion:** HyDE (Hypothetical Document Embeddings)
- **Table Processing:** TAPAS (Table Transformer) or Llama 2 with table prompting

Database:

- **Vector DB:** Chroma (lightweight) or FAISS (Facebook AI Similarity Search)
- **Document Store:** PostgreSQL (metadata, document paths)
- **Caching:** Redis (query caching, session management)

APIs:

- **SEC EDGAR API:** https://www.sec.gov/edgar/searchedgar/companysearch.html (free, no auth)
- **BSE India API:** Web scraping or GitHub datasets
- **REST API:** FastAPI endpoints for queries, document upload

Deployment:

- **Cloud:** AWS EC2 (t3.medium for backend, g4dn.xlarge for GPU inference) or GCP Compute Engine
- **Containerization:** Docker + Docker Compose
- **CI/CD:** GitHub Actions for automated deployment
- **Demo:** Streamlit (quick demo) or Gradio (LLM app demo)

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11. Main Modules

1. **Document Ingestion Module:**
    - SEC EDGAR API integration (automated 10-K/10-Q fetching)
    - BSE India scraping (annual reports)
    - PDF/HTML parsing (PyPDF2, BeautifulSoup)
2. **Document Preprocessing Module:**
    - Text extraction (PDF → text)
    - Table detection and HTML conversion (Camelot, Tabula)
    - Chunking (semantic chunking with sentence splitting)lib.uom
3. **Metadata Generation Module:**
    - LLM (Llama 2) generates chunk summaries

- NER (FinBERT) extracts company names, years, financial metrics
- Metadata storage (company, year, document_type, section, page_number)

4. **Embedding Generation Module:**
    - Dense embeddings (all-mpnet-base-v2)
    - Metadata-enriched embeddings (contextual chunks)
    - Vector storage (Chroma/FAISS)
5. **Query Processing Module:**
    - NER (FinBERT) for entity extraction
    - HyDE query expansion (LLM generates hypothetical answer)
    - Query decomposition (multi-hop reasoning agent)
6. **Retrieval Module:**
    - Dense search (cosine similarity on embeddings)
    - Sparse search (BM25 keyword matching)
    - Metadata filtering (company, year, document_type)
    - Hybrid fusion (weighted combination of dense + sparse scores)
7. **Reranking Module:**
    - Cross-encoder (ms-marco-T5-base) reranks top-50 → top-10
    - Custom metadata reranker (company/year match boost)
8. **Answer Generation Module:**
    - LLM (Llama 2/Mistral) generates answer from top-10 chunks
    - Citation extraction (page numbers, sections, tables)
    - Hallucination check (answer grounded in retrieved chunks)
9. **Caching Module:**
    - Redis cache for repeated queries
    - Embedding cache (avoid recomputing for same documents)
10. **Frontend Module:**

- React web app with query input
- Answer display with citations (clickable links to source documents)
- Financial chart visualization (revenue trends, etc.)

11. **Deployment Module:**
    - Docker containerization
    - AWS/GCP deployment scripts
    - Monitoring (Prometheus + Grafana for latency, accuracy tracking)

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12. Input and Output

Input:

- **User Query:** Natural language financial question (e.g., "What was Apple's Q3 2024 revenue?")
- **Optional Filters:** Company name, fiscal year, document type (10-K/10-Q)
- **Document Upload:** User can upload custom financial documents (PDF/HTML)

AI Processes:

1. **Query Understanding:** NER extracts entities (company, period, metric)
2. **Query Expansion:** HyDE generates hypothetical answer for better retrieval
3. **Retrieval:** Hybrid search (dense + sparse + metadata) finds relevant chunks
4. **Reranking:** Cross-encoder ranks chunks by relevance
5. **Answer Generation:** LLM synthesizes answer from chunks with citations

Output:

- **Answer:** Natural language response (e.g., "Apple's Q3 2024 revenue was $89.5B, up 8% YoY")
- **Citations:** Precise references (e.g., "[10-Q 2024, Page 12, Table 2]")
- **Confidence Score:** 0-1 score indicating answer confidence
- **Source Documents:** Clickable links to original 10-K/10-Q filings

- **Related Queries:** Suggested follow-up questions (e.g., "What was Apple's Q3 2023 revenue?")

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13. Evaluation Metrics

Retrieval Metrics:

- **Recall@10:** % of queries where correct evidence is in top-10 retrieved chunks (target: 92%+)arxiv
- **Recall@50:** % of queries where correct evidence is in top-50 (target: 95%+)
- **MRR (Mean Reciprocal Rank):** Average inverse rank of first correct result (target: 0.85+)
- **NDCG@10:** Normalized Discounted Cumulative Gain (ranking quality, target: 0.90+)

Generation Metrics:

- **Accuracy:** % of answers matching ground truth (target: 85%+ on FinanceBench)lib.uom
- **Faithfulness:** % of answer statements grounded in retrieved chunks (target: 95%+)
- **Completeness:** % of query aspects covered in answer (target: 90%+)
- **Hallucination Rate:** % of answer statements not in retrieved chunks (target: <5%)

System Metrics:

- **Latency:** P95 response time (target: <3 seconds per query)
- **Throughput:** Queries per second (target: 10+ QPS)
- **Caching Hit Rate:** % of queries served from cache (target: 30%+)

User Metrics:

- **User Satisfaction:** Survey score (1-5 Likert scale, target: 4.0+)
- **Citation Click-Through Rate:** % of users clicking citations (target: 40%+)

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14. Expected Improvement

Over Base Paper:lib.uom

Metric	Base Paper	Your Improvement	Gain
Accuracy (FinanceBench)	88%	90-92%	+2-4%
Recall@10	85%	92%+	+7%
Multi-Hop Queries	Not evaluated	80%+ accuracy	New capability
Table Handling	Text-only	Table-aware chunking	New capability
Latency (P95)	3-5 seconds	<3 seconds	40% faster
Citation Precision	Section-level	Page + table row/column	More precise
Indian Market Support	US-only	SEC + BSE/NSE	Geographic expansion

Key Improvements:

1. **88% → 90-92% accuracy** on FinanceBench via table-aware chunking + HyDE query expansion
2. **Multi-hop reasoning** (not in base paper) with 80%+ accuracy on complex queries
3. **40% latency reduction** via model distillation (MiniLM) + caching
4. **Table handling** (balance sheets, income statements) with 90%+ table cell retrieval accuracy
5. **Indian market extension** (BSE/NSE filings) for regional relevance

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15. Difficulty Level

Rating: Moderate to Advanced (3.5/5)

Why:

- **Moderate:** RAG pipeline is well-documented (LangChain, LlamaIndex tutorials)
- **Advanced:** Table-aware chunking, multi-hop reasoning, HyDE query expansion add complexity
- **Not Research-Level:** All components are available in open-source libraries (HuggingFace, LangChain)

Prerequisites:

- Python programming (FastAPI, PyTorch)
- Basic NLP knowledge (embeddings, transformers)
- Web development (React, REST APIs)
- No PhD-level AI knowledge required

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16. Implementation Feasibility

Dataset Availability: ⭐ ⭐ ⭐ ⭐ ⭐ (5/5)

- FinanceBench, FinDER: HuggingFace (instant download)
- SEC EDGAR: Public API (no auth, free)
- BSE/NSE: Public website (scraping or GitHub datasets)

Computing Requirements: ⭐ ⭐ ⭐ ⭐ (4/5)

- **Training:** None (use pre-trained models: Llama 2, all-mpnet-base-v2)
- **Inference:** GPU optional (Llama 2 7B runs on CPU, but GPU recommended for <3s latency)
- **Cloud Cost:** ~$50-100/month (AWS t3.medium + optional g4dn.xlarge for GPU)

Development Complexity: ⭐ ⭐ ⭐ (3/5)

- **RAG Pipeline:** LangChain/LlamaIndex simplifies implementation
- **Table Processing:** TAPAS or Llama 2 with table prompting (moderate complexity)
- **Multi-Hop Reasoning:** LLM agent for query decomposition (moderate)
- **Frontend:** React + Material-UI (standard web dev)

Deployment Difficulty: ⭐ ⭐ ⭐ ⭐ (4/5)

- **Docker:** Standard containerization
- **Cloud:** AWS EC2 or GCP (well-documented)
- **Demo:** Streamlit/Gradio (1-click deploy)

Estimated Development Time: 5-6 months

Timeline:

- **Month 1:** Literature review, setup environment, download datasets (FinanceBench, FinDER)
- **Month 2:** Implement baseline RAG (reproduce base paper 88% accuracy)lib.uom
- **Month 3:** Add table-aware chunking, metadata generation, hybrid search
- **Month 4:** Implement HyDE query expansion, cross-encoder reranking
- **Month 5:** Build React frontend, deploy on AWS/GCP
- **Month 6:** Testing, optimization (latency, accuracy), paper writing

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17. Publication Potential

Rating: High (4.5/5)

Why:

1. **Hot Topic:** RAG for financial QA is 2024-2025 cutting-edge (100+ papers in arXiv)computer+3
2. **Novel Contributions:** Table-aware chunking + multi-hop reasoning + Indian market extension are novel
3. **Strong Baselines:** FinanceBench, FinDER are standard benchmarks (easy comparison with base paper)arxiv
4. **Practical Deployment:** Working web app + AWS deployment adds real-world value
5. **Measurable Improvement:** 88% → 90-92% accuracy is publishable (even 2-4% gain is significant in RAG research)

Target Venues:

- **Conferences:**
    - **CIKM 2026** (ACM International Conference on Information and Knowledge Management) — FinSage was accepted herearxiv
    - **IEEE BigData 2026** (IEEE International Conference on Big Data) — MultiFinRAG was accepted herecomputer
    - **ACL Findings 2026** (Association for Computational Linguistics)

    - **IEEE ICDF2C 2026** (International Conference on Digital Forensics and Cyber Crime — fintech track)
- **Journals:**
    - **Expert Systems with Applications** (Elsevier, IF: 8.5)
    - **Decision Support Systems** (Elsevier, IF: 8.9)
    - **IEEE Transactions on Knowledge and Data Engineering** (IF: 8.9)
- **Preprint:**
    - **arXiv:** Fast publication (1-2 weeks), high visibility in AI community

Publication Strategy:

1. **Month 1-4:** Implement system, achieve 90%+ accuracy on FinanceBench
2. **Month 5:** Run ablation studies (table-aware chunking, HyDE, multi-hop reasoning contributions)
3. **Month 6:** Write paper (8-10 pages, IEEE/ACM format)
4. **Month 7:** Submit to CIKM 2026 or IEEE BigData 2026 (deadlines: May-June 2026)
5. **Backup:** arXiv preprint (immediate visibility)

Comparison Table

Project	AI Area	Real-World Impact	Base Paper Quality	Novelty	Difficulty	Dataset Availability	Publication Potential
FinRAG-Enhanced	NLP + RAG + LLMs	High (financial analysts, investors)	arXiv 2025, strong (88% accuracy)	High (table-aware + multi-hop + Indian market)	Moderate-Advanced	Excellent (FinanceBench, FinDER, SEC EDGAR)	High (CIKM, IEEE BigData, arXiv)